# Multi-Camera Vehicle Tracking Based Based on the ELECTRICITY Framework


Presenter

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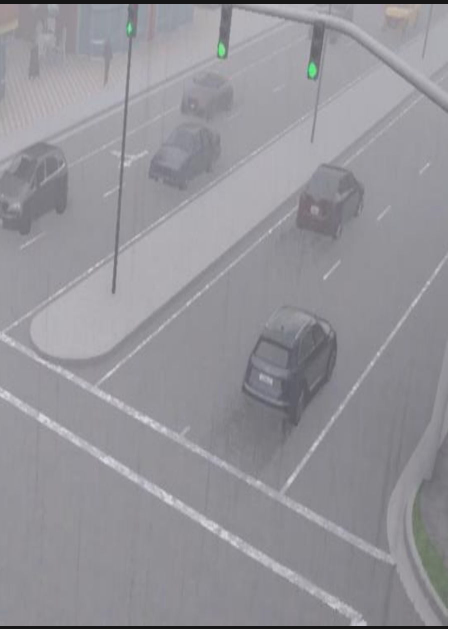
# What Is MCMT Tracking

## Definition of MCMT

Multi-Camera Multi-Target (MCMT) tracking is essential for intelligent transportation systems. It involves identifying vehicles across multiple non-overlapping camera views, ensuring consistent tracking despite changes in lighting, pose, and occlusions.

## Challenges of MCMT

Unlike single-camera tracking, MCMT faces significant challenges due to large spatiotemporal spatiotemporal gaps between camera views. Vehicles can appear drastically different across across cameras, making identity consistency a a complex task.

Real Data Limitations

Manual annotation of real multi-camera footage is expensive and time-consuming due to privacy constraints and the difficulty of synchronizing identities across cameras.

# Synthetic vs Real Data

Advantages of Synthetic Data

Simulators like CARLA can generate unlimited, perfectly labeled synthetic synthetic video with controllable parameters such as weather, density, and density, and viewpoint. This provides a cost-effective alternative for for training models.

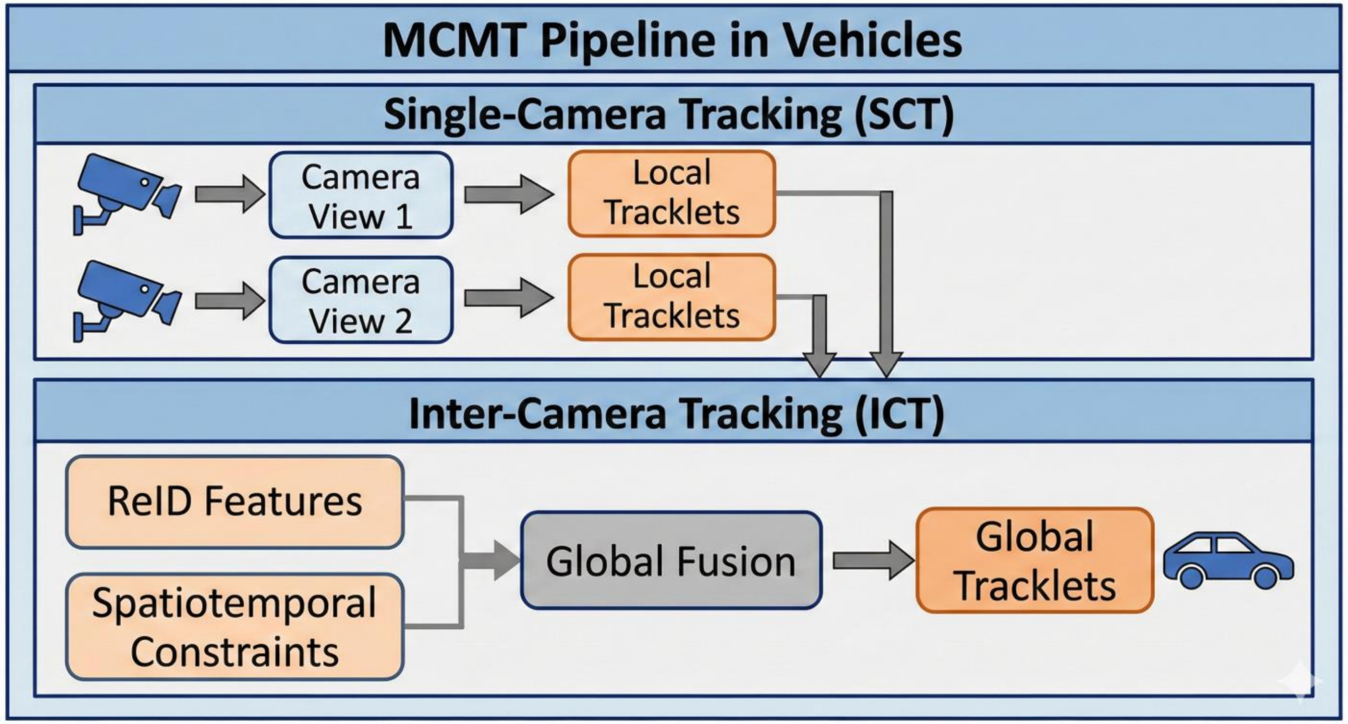
Domain Gap Issue

However, models trained on synthetic data often struggle to generalize to generalize to real-world footage due to differences in texture, lighting, and lighting, and noise, creating a significant domain gap.

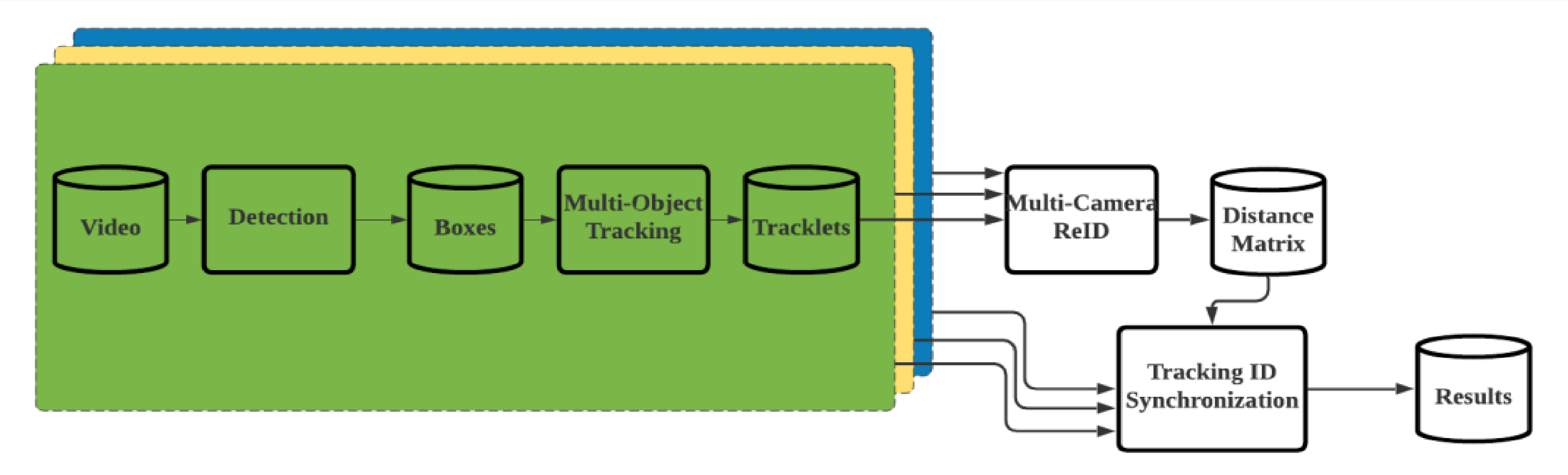
# Two-Stage MCMT Pipeline

## Pipeline Stages

The MCMT pipeline consists of two stages: Single-Single-Camera Tracking (SCT) and Inter-Camera Tracking Tracking (ICT). SCT forms local tracklets within each each camera view, while ICT fuses these tracklets globally using ReID features and spatiotemporal constraints.



# ELECTRICITY Model Design




## ELECTRICITY Overview

The ELECTRICITY model formulates tracking as an efficient energy minimization problem. It constructs a constructs a tracklet graph where nodes are single-single-camera trajectories and edges represent ReID ReID similarity and spatiotemporal feasibility.

## Efficiency and Performance

This approach allows for real-time, online association association without the need for computationally expensive offline graph-cut methods, making it suitable suitable for large-scale multi-camera tracking applications.

# Vehicle Detection by Mask R-CNN CNN

## 01

### Mask R-CNN Function Function

Mask R-CNN detects vehicles in each frame, providing bounding boxes, masks, and confidence scores. It focuses on three categories: cars, buses, and trucks, treating them as vehicles for subsequent processing.

## 02

### Inter-class NMS

To handle duplicate detections, detections, Inter-class Non-maximum Suppression (NMS) is (NMS) is applied. This cross-class class filtering ensures accurate accurate detection by eliminating redundant bounding bounding boxes based on IoU IoU and confidence scores.

## 03

### Output for Tracking

The clean detection results from from Mask R-CNN serve as input input for single-camera multi-multi-object tracking, providing providing robust initial data for for the MCMT pipeline.

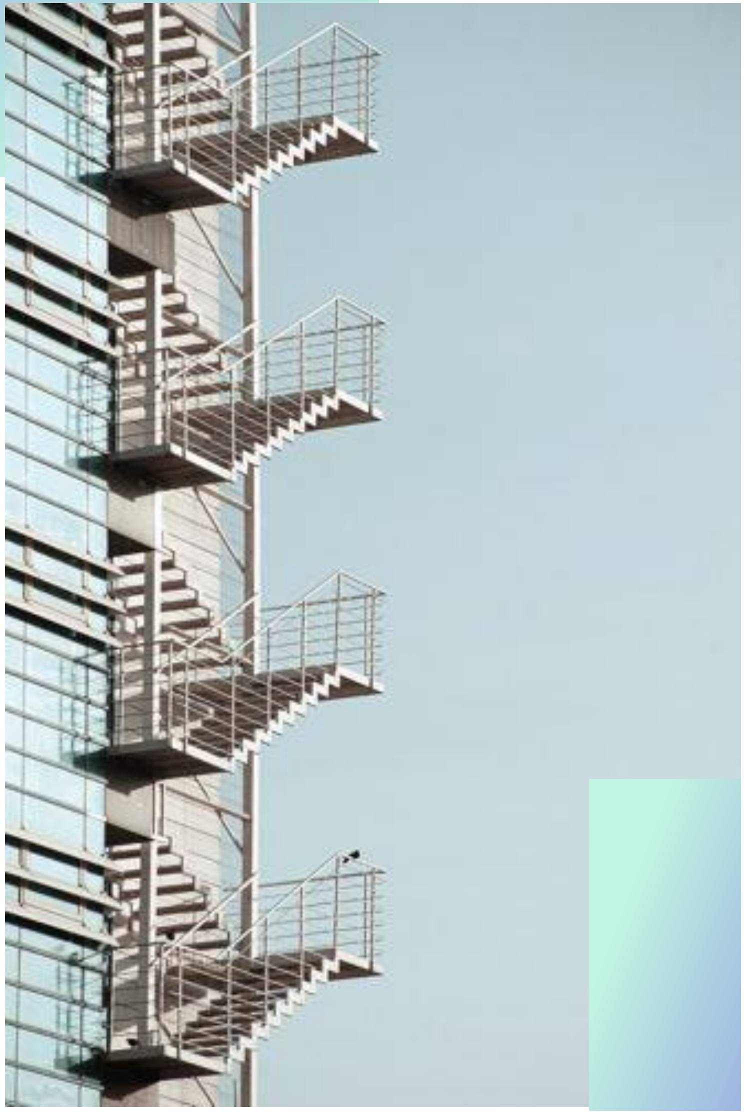
# DeepSORT Inside Each View

## DeepSORT Mechanism

DeepSORT links vehicle detections into continuous tracklets within each camera view. It uses Kalman filters for motion prediction and appearance features from Mask R-CNN for association.

## Hierarchical Matching

The matching process first compares confirmed trajectories trajectories using appearance appearance features. If unmatched, it falls back to Intersection over Union (IoU) of of bounding boxes, ensuring robust tracking even with occlusions.



# ReID Network Backbone

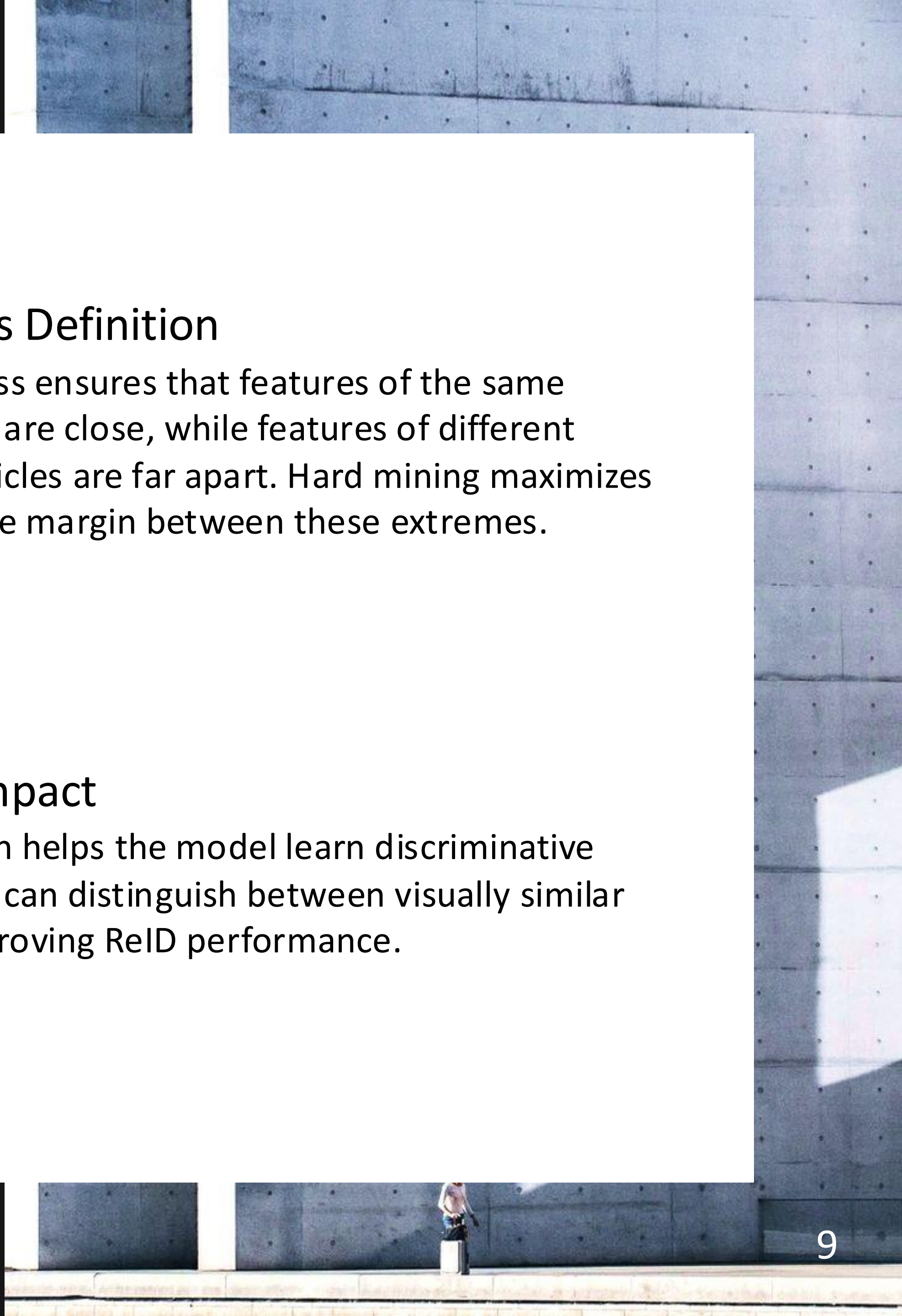
## ReID Architecture

The ReID network uses a ResNet-101 backbone to backbone to extract 2048-D appearance embeddings embeddings from vehicle crops. It is trained with an with an aggregated loss combining cross-entropy and entropy and triplet loss with hard mining.

# Hard Triplet Mining Mining Explained

## Hard Positive and Negative

Hard positive samples are the most dissimilar images of the same vehicle, while hard negative samples are the most similar images from different vehicles. This forces the model to focus on subtle discriminative details.



## Triplet Loss Definition

The triplet loss ensures that features of the same same vehicle are close, while features of different different vehicles are far apart. Hard mining maximizes maximizes the margin between these extremes.

## Training Impact

This approach helps the model learn discriminative features that can distinguish between visually similar vehicles, improving ReID performance.

# Synthehicle Dataset Portrait

## Synthehicle Features

The Synthehicle dataset offers 4 virtual towns with day, night, dawn, and rain conditions. It provides perfectly labeled multi-camera sequences with synchronized identities and realistic traffic density.

## Training Benefits

Synthehicle's unlimited synthetic data is ideal for training due to its perfect annotations and controllable parameters, making it a valuable resource for large-scale MCMT benchmarking.

# CityFlow Real-World Benchmark

## 01

### CityFlow Overview

CityFlow contains 5-camera Scene S01 for training and 23-camera validation splits across S02 and S05. It features real-world intersections with manual identity annotations.

## 02

### Real-World Challenges

The dataset captures complex lighting, heavy occlusion, and similar-looking vehicles, making it a making it a challenging testbed for for evaluating sim-to-real transfer transfer performance.

## 03

### Evaluation Importance

CityFlow serves as the authoritative authoritative benchmark for assessing how well models trained trained on synthetic data can generalize to real-world surveillance surveillance scenarios.

# ReID on Synthehicle

## Synthehicle Training

Training the ReID model exclusively on Synthehicle data achieves 77.2% mAP and 64.1% Rank-1 within 17 epochs. This indicates excellent memorization of synthetic textures but limited generalization to real-world scenarios.

## CityFlow Fine-Tuning

Fine-tuning the ReID model on CityFlow data data reaches 100% Rank-1 and 100% mAP by by epoch 13. However, this high performance is performance is due to overfitting to specific specific camera views and background biases. biases.

## Overfitting Concerns

The validation set constructed from training data training data reveals overfitting issues. True True generalization requires evaluation across across diverse scenes and weather conditions. conditions.

# ReID on CityFlow

# ELECTRICITY

# Overview of ELECTRICITY

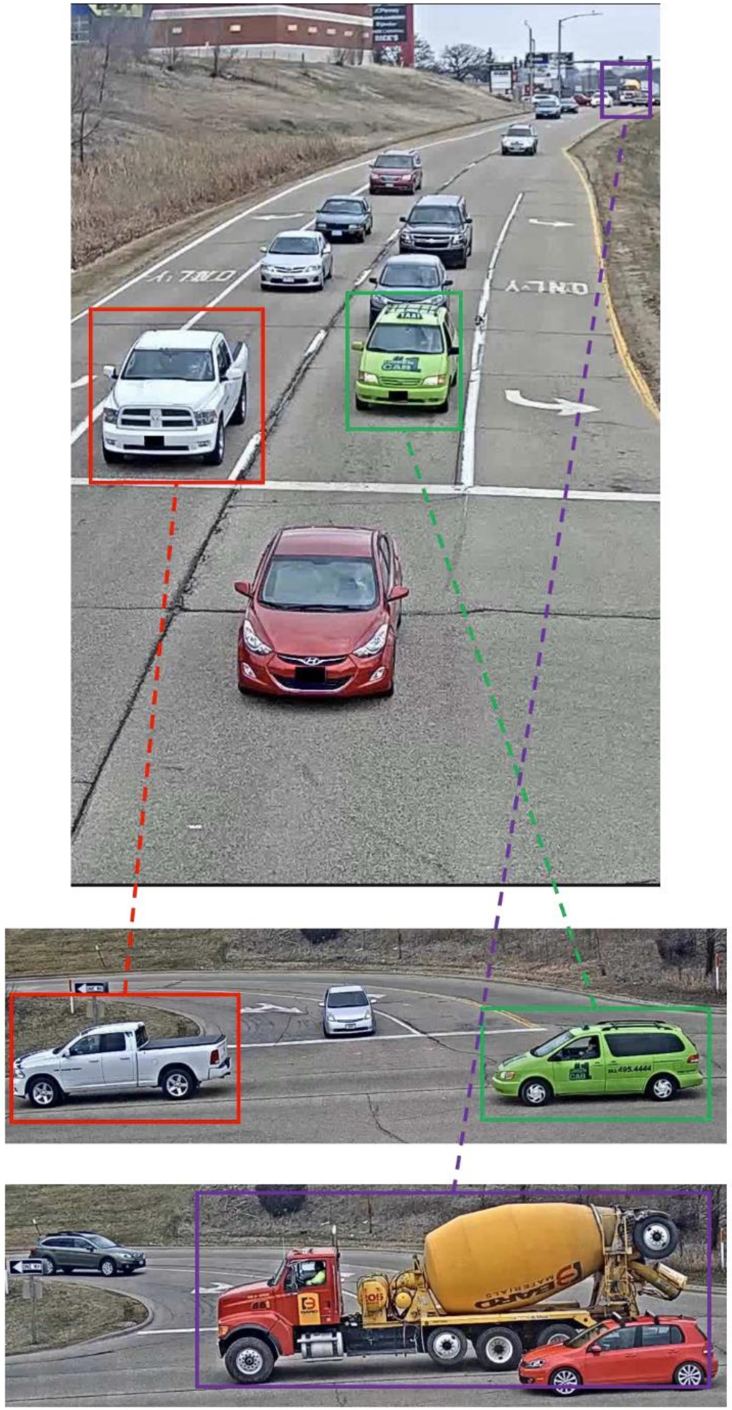
First, ELECTRICITY detects vehicles using a pretrained model, Mask R-CNN. The result is frames of cropped vehicles. This is fed into the single-camera tracking model, DeepSORT, which outputs tracklets that contain tracking information. Next, the single-camera tracklets are fed into the vehicle reidentification model, ReID. This pairs tracklets to give each vehicle a single ID per camera. Finally, the tracklets are correlated to assign a global ID to each vehicle.

Detection

Multi-target Single-camera Tracking

Vehicle Reidentification

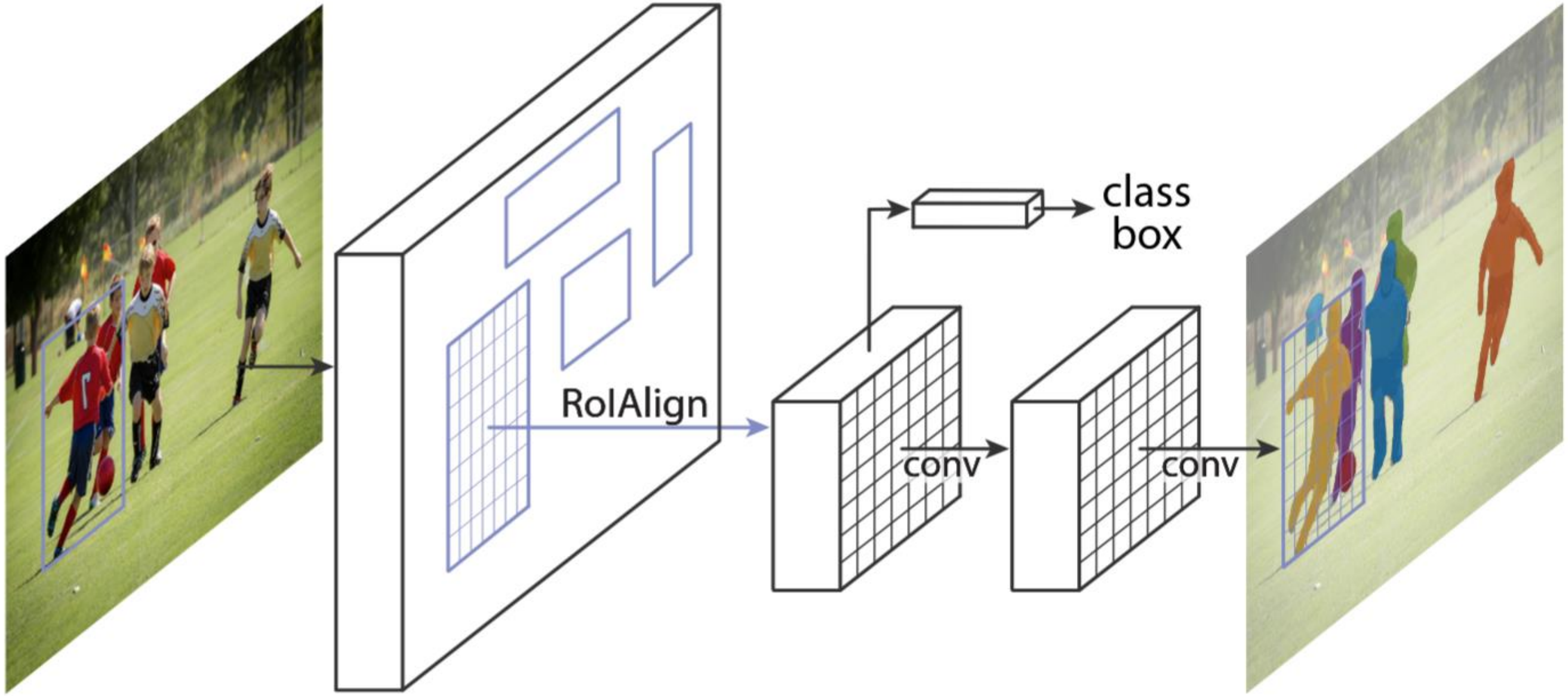
MTMCT



# Detection

## Mask R-CNN

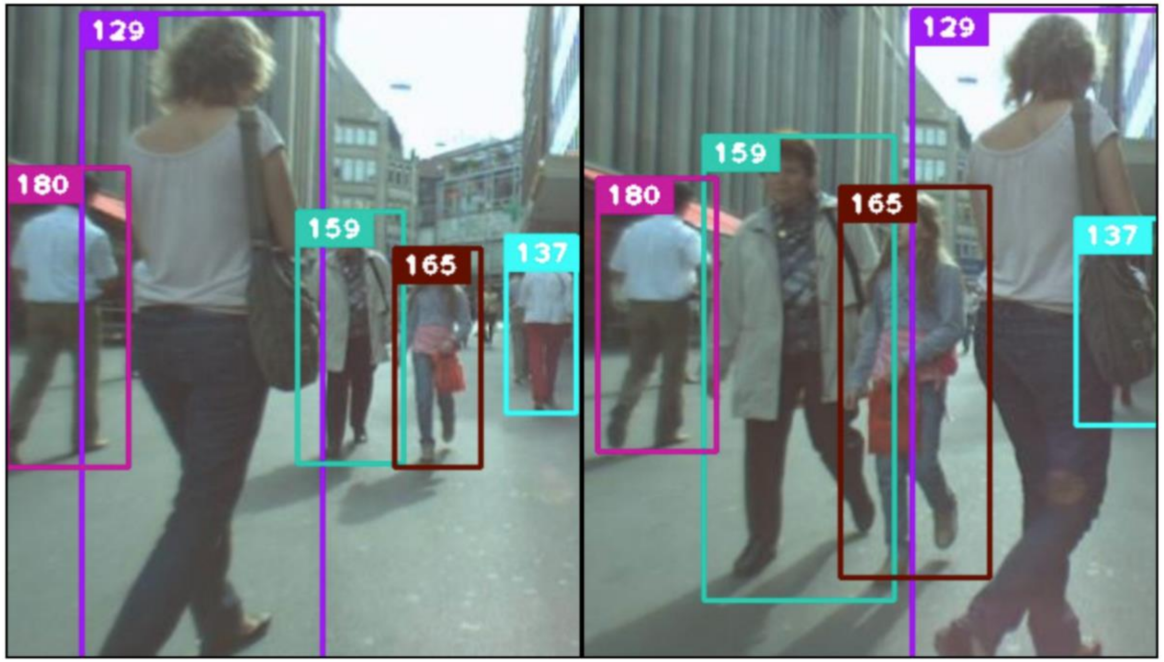
A deep convolutional neural network that incorporates the RoI Align mechanism to generate candidate target regions from the image. Its multi-head structure allows it to simultaneously output target category information, confidence scores, bounding box coordinates, and pixel-level segmentation masks.



# Multi-target Single-camera Tracking

## DeepSORT

For each vehicle target, the system uses a Kalman filter to build its motion state model and predicts the target’s position at the next moment under the assumption of uniform motion. The ELECTRICITY model first uses appearance features and spatial constraints to match confirmed trajectories with the detection results of the current frame. For targets that still do not match, a second matching is performed using the Intersection over Union (IoU) of bounding boxes. Finally, detection results that still do not match are initialized as new trajectories

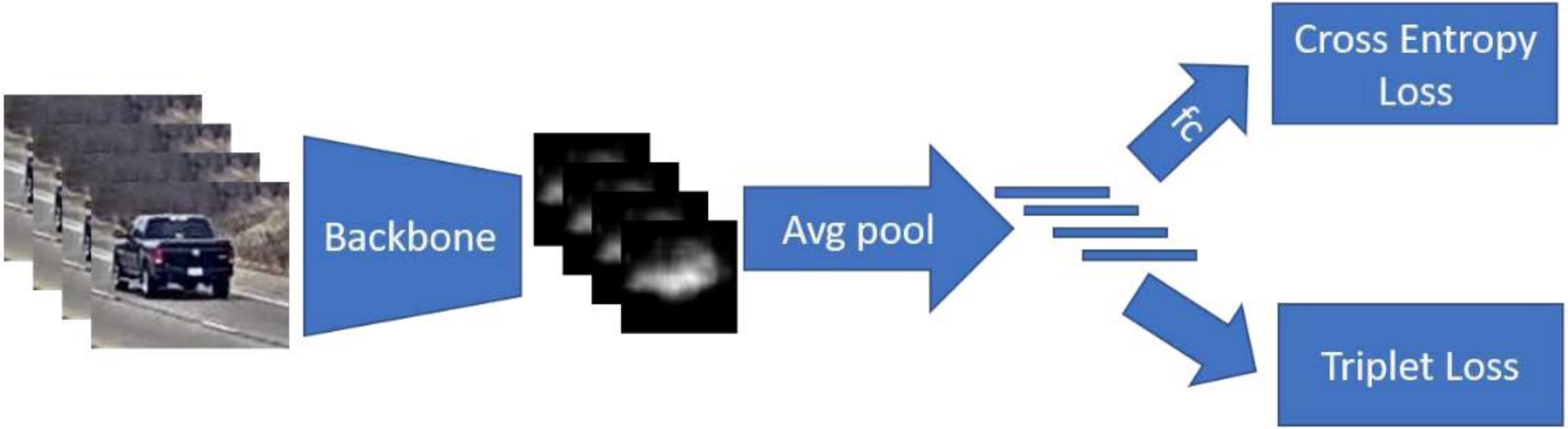



Exemplary output from the DeepSORT paper, pedestrians instead of vehicles

# Vehicle Reidentification

## ReID

During training, images from different vehicle identities are sampled to form a batch. Each image is sent into a deep neural network f(·) to extract an appearance feature vector. The ReID network is optimized using an aggregation loss, which combines triplet loss and cross-



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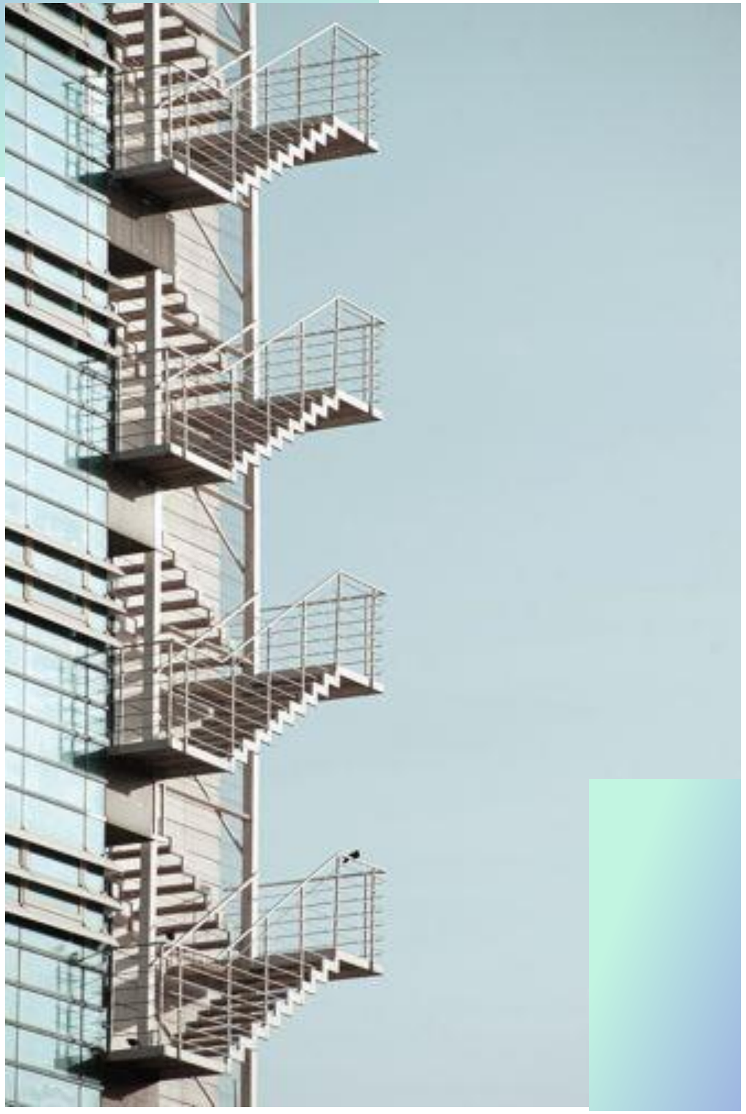


# MTMCT

## Mask R-CNN

The model first calculates the distance between appearance features of each pair of tracks from different cameras, combined with the location relationship between cameras. Since vehicles in real-world scenarios can only pass adjacent cameras sequentially, the model only associates logically adjacent cameras.

A MTMCT visualization from the ELECTRICITY paper. Each row represents a tracklet. Red means false positive. Yellow means false negative.

# Experiments

# Reproducing Synthehicle Results

## MTMCT Results for Syntheicle data

First we used the pretrained ELECTRICITY model on the pre-generated data provided by syntheicle. This way, we have preliminary results generated under the same conditions as the syntheicle MTMCT results.

# MTMCT results

Town01-N-night (pretrained)

## IDF1: 0.7485

IDP: 2.0100

IDR: 2.0100

Recall: 0.5502

MOTA: -1.9537

MOTP: 0.2550

# Training for Reidentification

## Training on Syntheicle data

We fine-tuned the same pretrained model from the previous experiment on more syntheicle data using the ReID module of the ELECTRICITY framework. All of the training data this model has seen is synthetic

## Training on CityFlow data

We fine-tuned the previous model again, but this time with CityFlow data using the ReID module of the ELECTRICITY framework. The performance of this model should represent the performance of a model trained on real-world data.

# City Flow Preprocessing

## Problem

The CityFlow dataset is oranized into a structure that is not compatible with the ELECTRICITY framework. The CityFlow format is a sequence of images that form a video and a groud truth txt file for the bounding boxes.

## Preprocessing Solution

We cropped all vehicles out of each image and stored them in an image format that follows ELECTRICITY's naming convention.

This is the ELECTRICITY instance that was fine-tuned on syntheicle data.

# Fine-Tuned Syntheicle Results

Town01-N-night (fine-tuned)

## IDF1: 0.3968

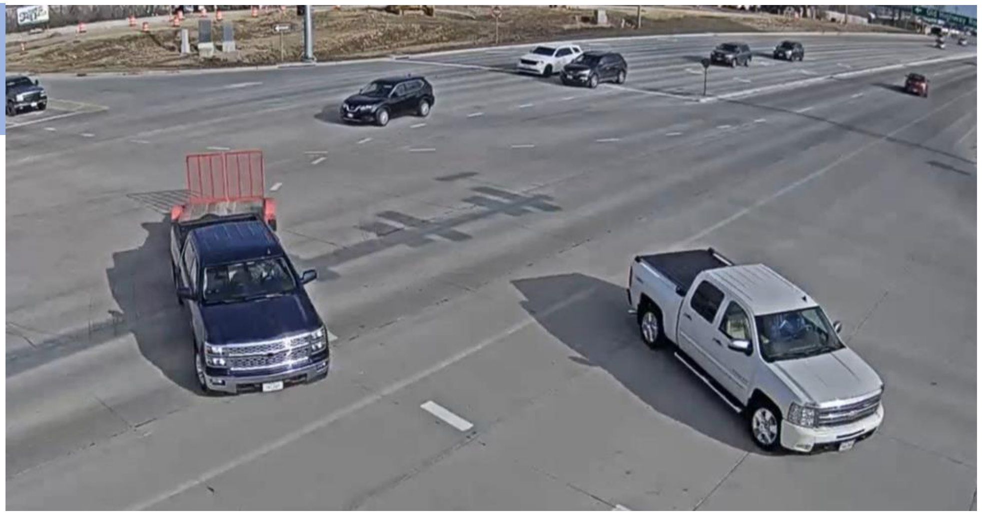
IDP: 1.3499

IDR: 0.3533

Recall: 0.2197

MOTA: -0.3442

MOTP: 0.2469



Cam6 to Cam9 in S02 (best)

IDF1%: 19.25

IDP%: 17.33

IDR%: 21.65

This is the ELECTRICITY instance that was fine-tuned on CityFlow data.

# Fine-Tuned CityFlow Results

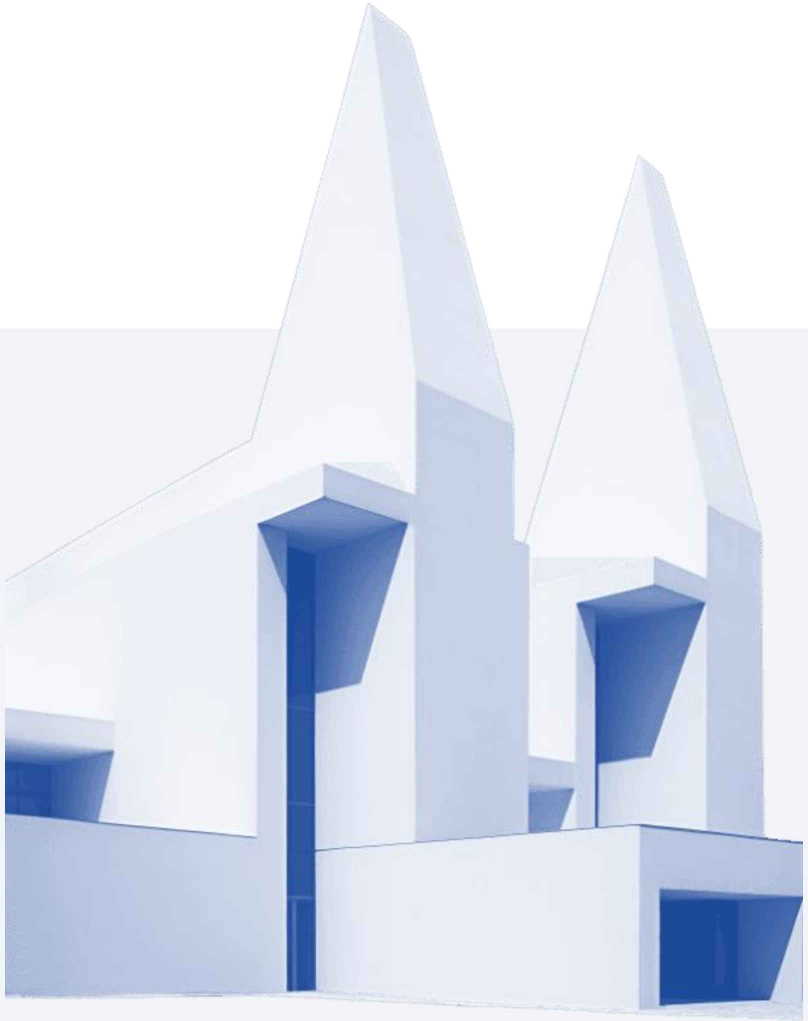
# Metrics IDF1 IDP IDR

## IDP and IDR Anomalies

IDP and IDR values can exceed 1 due to due to multi-camera stitching, causing causing anomalies in evaluation. IDF1 is IDF1 is more reliable for assessing global global identity consistency across cameras.

## IDF1 Metric

IDF1 balances identity precision and recall, providing a robust measure of of tracking performance. It remains unaffected by multi-camera stitching stitching artifacts that inflate IDP and IDR. and IDR.



## Performance Comparison

The model achieves 0.7485 IDF1 on on Synthehicle's night scene and 19.25% IDF1 on CityFlow's S02, highlighting the persistent domain domain gap between synthetic and and real-world data.

01

## Domain Gap Causes

The domain gap arises from differences in texture, lighting, and noise between synthetic and real data. Synthetic textures lack real sensor noise and weather reflections, causing ReID features to overfit.

02

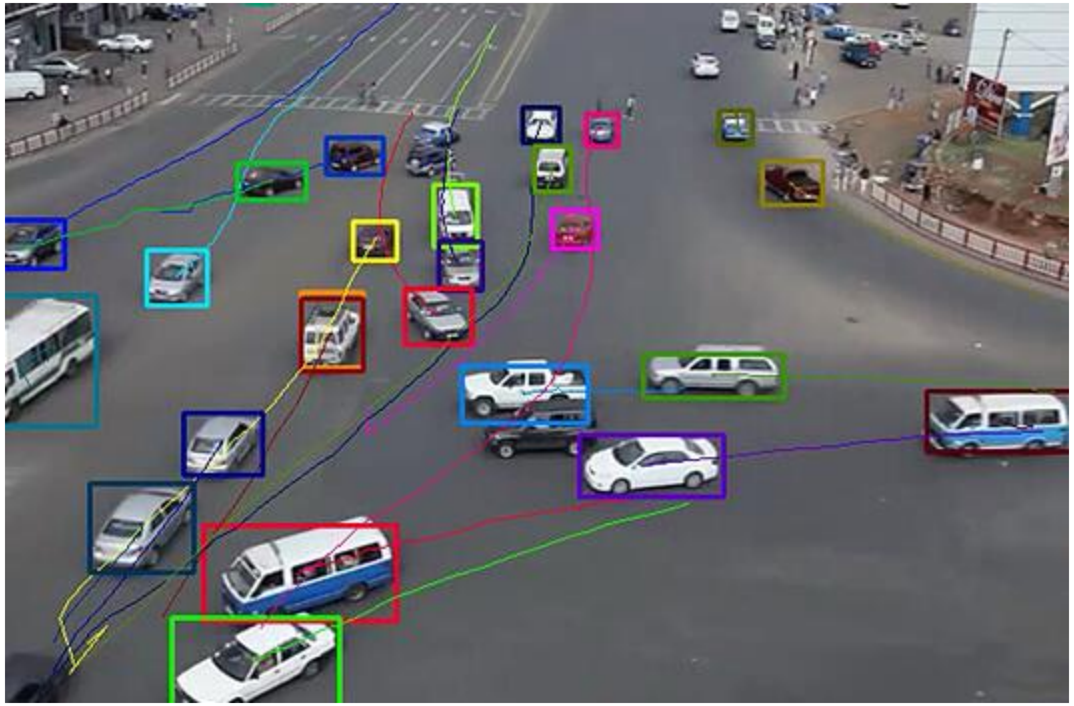
## Evaluation Sensitivity

Temporal stitching evaluation further accumulates identity switches, leading to leading to inflated IDP values and negative MOTA. This highlights the need for need for more robust evaluation protocols.

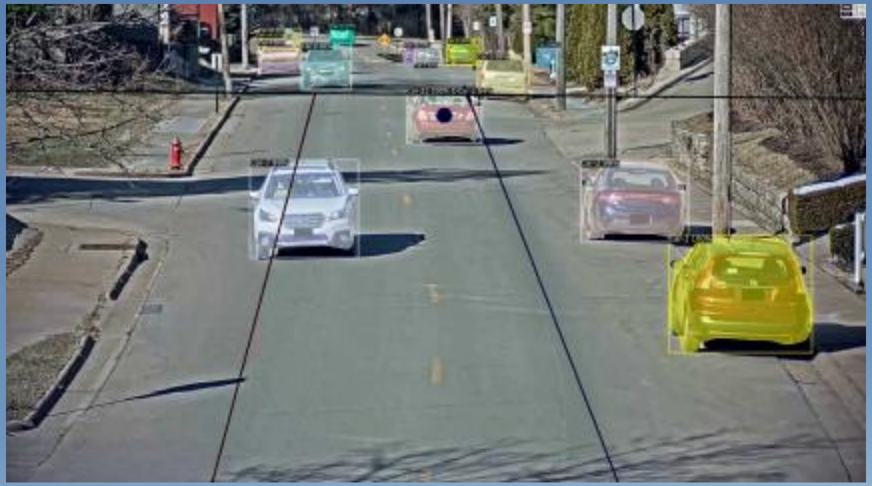
# Sim-to-Real Gap Analysis

CHAPTER

# Conclusion & Future Work



# Key Takeaways



## ELECTRICITY Limitations

ELECTRICITY achieves efficient online MCMT, but ReID generalization remains limited by the domain gap. Synthetic data provides cheap annotation but requires adaptation to real-world conditions.

## Future Directions

Future work includes expanding training to multi-weather weather real sequences, integrating domain adaptation adaptation techniques, and refining evaluation protocols to protocols to prevent metric inflation.

# Future Work Roadmap

### Diverse Training Data

Expand training to include multi-weather real sequences to improve model robustness and generalization.

### Domain Adaptation

Integrate unsupervised domain adaptation techniques to bridge the the gap between synthetic and real real data.

### Continuous Learning

Develop a self-adapting tracker that that continuously learns from incoming incoming surveillance streams without without manual re-annotation.

# Contributors and Roles

## Yuexin Chen

Yuexin Chen handled dataset preparation, model implementation, training, and evaluation pipelines, ensuring the project's technical foundation.

## Samanwita Das

Samanwita Das authored the introduction, background, and related related work sections and generated generated CARLA sample data, contributing to the project's documentation and data generation. generation.

## Luke Waind

Luke Waind set up GPU server that we could all connect to and work, provided supplemental help with environment setup and evaluation for ELECTRICITY, and helped with report editing.

# References and Thanks

## Acknowledgements

This work builds on ELECTRICITY, Synthehicle, CityFlow, CARLA, Mask R-CNN, DeepSORT, and MOTChallenge metrics. We thank the open-source communities and dataset authors for enabling reproducible research in multi-camera vehicle tracking and synthetic data generation.

# THANK YOU